A ten-variable Diophantine candidate

Strictly positive integer unknowns; exact degree 750; 80 evaluator operations.

The only unknowns are $a, b, c, d, e, f, g, h, i, j \in \mathbb{Z}_{>0}$. Each $\texttt{r01}, \ldots, \texttt{r80}$ below is an abbreviation, not an additional unknown. Evaluate the rows in order using ordinary integer arithmetic; subtraction is integer subtraction, and intermediate values may be negative. The equation is $\boxed{\texttt{r80} = 0}$.

$\texttt{r01} = \beta \cdot a$
$\texttt{r02} = \texttt{r01} + 1$
$\texttt{r03} = \texttt{r02} \cdot \texttt{r02}$
$\texttt{r04} = \texttt{r03} \cdot \texttt{r02}$
$\texttt{r05} = \texttt{r04} \cdot g$
$\texttt{r06} = \texttt{r02} - b$
$\texttt{r07} = \texttt{r06} - c$
$\texttt{r08} = f \cdot \texttt{r02}$
$\texttt{r09} = i + j$
$\texttt{r10} = \texttt{r09} \cdot \texttt{r05}$
$\texttt{r11} = \texttt{r10} + i$
$\texttt{r12} = \texttt{r03} + 1$
$\texttt{r13} = \texttt{r08} + 1$
$\texttt{r14} = \texttt{r12} \cdot \texttt{r05}$
$\texttt{r15} = \texttt{r14} \cdot \texttt{r13}$
$\texttt{r16} = 4 \cdot \texttt{r15}$
$\texttt{r17} = \texttt{r16} - 1$
$\texttt{r18} = \texttt{r15} + 2$
$\texttt{r19} = \texttt{r14} \cdot \texttt{r14}$
$\texttt{r20} = \texttt{r08} \cdot \texttt{r19}$
$\texttt{r21} = \texttt{r20} \cdot \texttt{r09}$
$\texttt{r22} = \texttt{r21} + \texttt{r09}$
$\texttt{r23} = \texttt{r21} \cdot \texttt{r22}$
$\texttt{r24} = \texttt{r23} + 1$
$\texttt{r25} = \texttt{r01} \cdot \texttt{r06}$
$\texttt{r26} = \texttt{r25} + c$
$\texttt{r27} = \texttt{r03} - 1$
$\texttt{r28} = \texttt{r26} \cdot \texttt{r27}$
$\texttt{r29} = \texttt{r02} \cdot \nu$
$\texttt{r30} = \mu + \texttt{r29}$
$\texttt{r31} = \texttt{r30} \cdot a$
$\texttt{r32} = \texttt{r28} + \texttt{r31}$
$\texttt{r33} = \texttt{r09} - \texttt{r32}$
$\texttt{r34} = \texttt{r33} - 1$
$\texttt{r35} = \kappa + f$
$\texttt{r36} = \texttt{r35} \cdot \texttt{r07}$
$\texttt{r37} = \texttt{r36} - b$
$\texttt{r38} = \texttt{r11} \cdot \texttt{r11}$
$\texttt{r39} = d \cdot \texttt{r38}$
$\texttt{r40} = \texttt{r39} \cdot \texttt{r24}$
$\texttt{r41} = \texttt{r15} \cdot \texttt{r18}$
$\texttt{r42} = \texttt{r41} \cdot \texttt{r40}$
$\texttt{r43} = \texttt{r42} \cdot \texttt{r39}$
$\texttt{r44} = \texttt{r43} + 1$
$\texttt{r45} = \texttt{r43} \cdot \texttt{r44}$
$\texttt{r46} = 4 \cdot \texttt{r45}$
$\texttt{r47} = \texttt{r46} + 1$
$\texttt{r48} = \texttt{r41} \cdot \texttt{r46}$
$\texttt{r49} = \texttt{r11} \cdot e$
$\texttt{r50} = \texttt{r49} - \texttt{r32}$
$\texttt{r51} = \texttt{r47} \cdot \texttt{r50}$
$\texttt{r52} = \texttt{r51} - \texttt{r11}$
$\texttt{r53} = \texttt{r52} \cdot \texttt{r52}$
$\texttt{r54} = \texttt{r12} \cdot \texttt{r12}$
$\texttt{r55} = \texttt{r01} \cdot \texttt{r54}$
$\texttt{r56} = \texttt{r53} \cdot \texttt{r37}$
$\texttt{r57} = h - \texttt{r56}$
$\texttt{r58} = \texttt{r17} \cdot \texttt{r57}$
$\texttt{r59} = 3 \cdot \texttt{r11}$
$\texttt{r60} = 2 \cdot \texttt{r08}$
$\texttt{r61} = \texttt{r59} - \texttt{r60}$
$\texttt{r62} = \texttt{r08} \cdot \texttt{r61}$
$\texttt{r63} = \texttt{r62} + 2$
$\texttt{r64} = \texttt{r58} + \texttt{r63}$
$\texttt{r65} = \texttt{r55} \cdot \texttt{r64}$
$\texttt{r66} = \texttt{r54} \cdot \texttt{r37}$
$\texttt{r67} = \texttt{r01} \cdot \texttt{r34}$
$\texttt{r68} = \texttt{r66} + \texttt{r67}$
$\texttt{r69} = \texttt{r17} \cdot \texttt{r68}$
$\texttt{r70} = \texttt{r65} - \texttt{r69}$
$\texttt{r71} = \texttt{r55} \cdot \texttt{r17}$
$\texttt{r72} = \texttt{r71} \cdot \texttt{r71}$
$\texttt{r73} = \texttt{r24} \cdot \texttt{r44}$
$\texttt{r74} = \texttt{r70} \cdot \texttt{r70}$
$\texttt{r75} = \texttt{r72} \cdot \texttt{r73}$
$\texttt{r76} = \texttt{r75} \cdot \texttt{r53}$
$\texttt{r77} = \texttt{r74} - \texttt{r76}$
$\texttt{r78} = \texttt{r48} \cdot \texttt{r77}$
$\texttt{r79} = \texttt{r75} - \texttt{r74}$
$\texttt{r80} = \texttt{r78} + \texttt{r79}$

The four fixed integers. All indices in the following formulas are locally bound indices for constants, not unknowns. Set $\rho = 2^{67}$ and $\beta = \rho^{8807} - 1$. Let the fixed set and ordered list be

$$\begin{aligned}\mathcal{S} = \big(&\{0,\dots,942\} \cup \{983,\dots,992\} \cup \{1003,\dots,1011\} \cup \{1013,\dots,1031\} \\ &\cup \{1063, 1064, 1095, 2195, 3296\}\big) \setminus \{1\},\\ \lambda = (&1023, 1024, 1025, 1026, 1027, 1028, 1029, 1030, 1013, 1014,\\ &1015, 1016, 1017, 1018, 1019, 1020, 1021, 1003, 1004, 1005,\\ &1006, 1007, 1008, 1009, 1010, 983, 984, 985, 986, 987,\\ &988, 989, 990, 991, 1031, 1022, 1011, 992, 1095, 2195,\\ &3296).\end{aligned}$$

The list λ has 41 entries, numbered from zero. The fixed list σ has 943 entries, also numbered from zero, and is printed in decimal below and on the following pages. Then

$$\begin{aligned}\mu &= \beta - \sum_{u\in\mathcal{S}} \rho^u - 2\rho^{1063},\\ \nu &= \beta + 4 + \sum_{u=1013}^{1022} \rho^u + \sum_{u=1065}^{1073} \rho^u + \rho^{2195} + \rho^{3296}\\ &\quad + (2^{51}-2)(\rho^{4393} + \rho^{5495}) + (2^{65}-1)\sum_{u=8792}^{8804} \rho^u + (2^{19}-1)\rho^{8805},\\ \kappa &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849}\\ &\quad + \sum_{u=0}^{942} \sigma_u(\rho^{4393-u} + \rho^{5495-u}) + \sum_{u=0}^{40} 2^{10+u}(\rho^{4393-\lambda_u} + \rho^{5495-\lambda_u}).\end{aligned}$$

Decimal selector list. Read from left to right, top to bottom; each 14-digit block is one decimal integer. Leading zeroes do not change its value. There are exactly 943 blocks.

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